

DUAL-CORE FIRMWARE & HOST SOFTWARE

STM32H747 MotionIPC Architecture, Telemetry, and Characterization

Customer-facing technical report
Revision 2.3 • 3 August 2026

Prepared by	Herder Elektronische Systemen
Primary subject	STM32H747 dual-core firmware, inter-core communications, and host software
Evidence basis	MotionIPC source package, motion firmware, GUI video, logs, slides, plots, and technical reports
Distribution	Customer-facing; selected public source excerpts included; complete development archives excluded

DOCUMENT EMPHASIS

This revision makes the dual-core firmware architecture the primary engineering result. Mechanical and test-platform material is included only where it explains the target workload or validates the software stack.

1. Executive Summary

The system uses an STM32H747 heterogeneous dual-core microcontroller to separate high-rate motion data movement from low-rate control and coordination. A 400 MHz Cortex-M7 acts as the OpenAMP master and shared-ring producer; a 200 MHz Cortex-M4 acts as the OpenAMP remote and shared-ring consumer. The design combines RPMMsg control messages with a custom AXI-SRAM ring buffer so motion records do not compete with command, heartbeat, fault, and status traffic.

STM32H747 Dual-Core Motion Architecture

Low-rate coordination is separated from high-rate shared-memory data movement

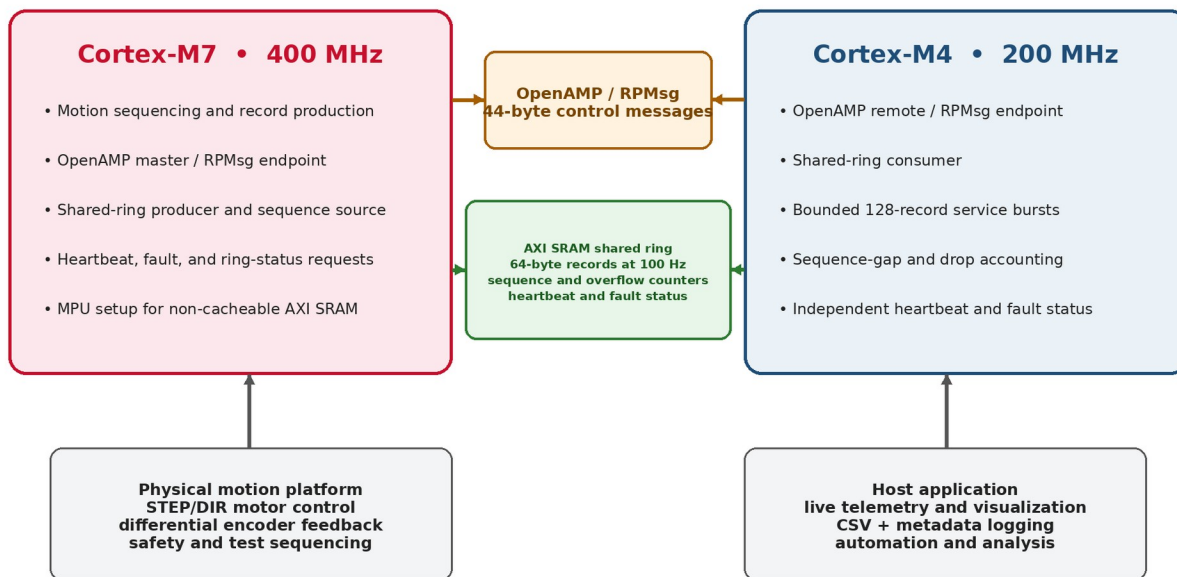


Figure 1. Dual-core architecture: OpenAMP/RPMMsg control plane and AXI-SRAM high-rate data plane.

KEY ENGINEERING RESULT

The firmware is not a single-core application placed on a dual-core board. It defines explicit ownership, memory regions, protocol messages, producer/consumer synchronization, cache behavior, sequence accounting, and per-core health reporting.

2. Verified Architecture Specifications

Verified Dual-Core Firmware Specifications

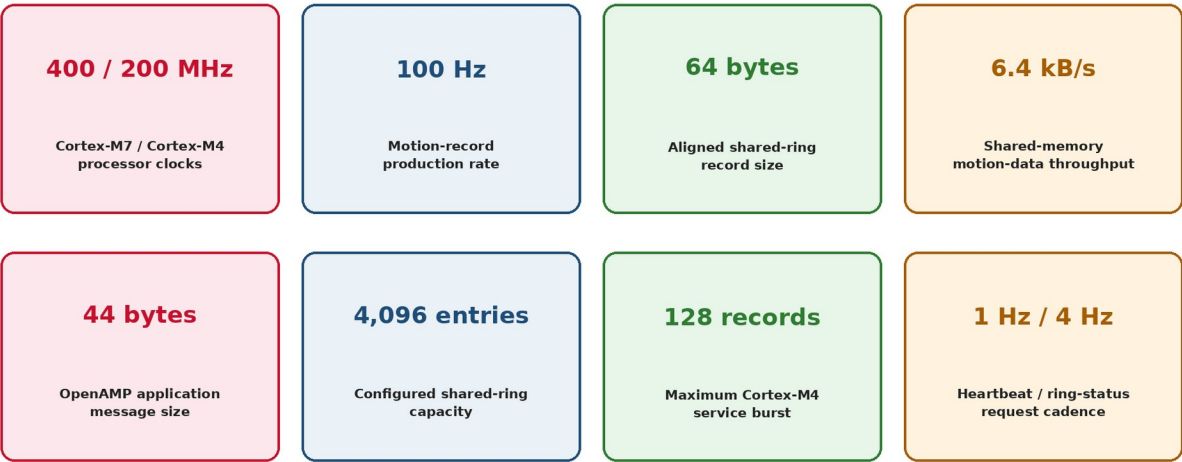


Figure 2. Source-verified processor, IPC, buffering, and service-cadence specifications.

Item	Verified implementation
Processor	STM32H747: 400 MHz Cortex-M7 + 200 MHz Cortex-M4
Control IPC	OpenAMP/RPMsg service named motion_ipc
High-rate IPC	Custom producer/consumer ring in AXI SRAM
Motion record	64-byte, 32-byte-aligned MotionRawSample
Application message	44-byte MotionIpcMessage with 8 × 32-bit arguments
Production rate	100 Hz in the supplied MotionIPC integration revision
Ring allocation	256 KiB at 0x24000000
Configured capacity	4,096 entries; engineering review identifies a ~64-byte allocation overrun
Consumer service	Bounded bursts of up to 128 records per CM4 pass
Health/status	1 Hz core heartbeats and 4 Hz ring-status requests

3. Core Partitioning and Boot Sequence

3.1 Cortex-M7 responsibilities

- Configures the MPU and cache policy before enabling the high-rate shared-memory path.
- Boots and releases the Cortex-M4 through hardware semaphore HSEM0.
- Initializes the AXI-SRAM ring as producer and OpenAMP as RPMsg master.
- Produces timestamped motion records and owns the monotonically increasing record sequence.
- Sends heartbeat and ring-status requests and receives fault/status replies.

3.2 Cortex-M4 responsibilities

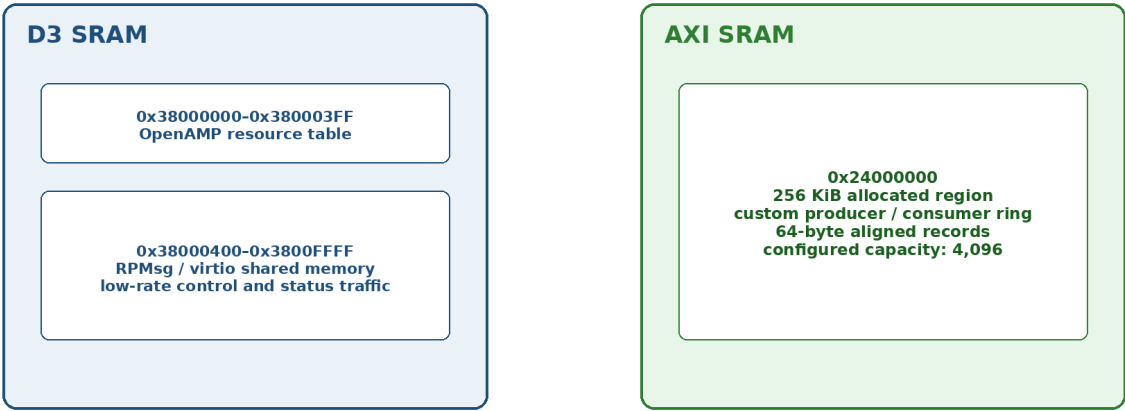
- Wakes from STOP mode in response to HSEM signaling and initializes as the OpenAMP remote.
- Consumes motion records from the shared ring without routing each record through RPMsg.
- Processes up to 128 records per service pass so OpenAMP polling remains responsive.
- Counts samples, sequence gaps, dropped records, and last-consumed sequence.
- Reports independent heartbeat, ring status, and fault information to the M7.

3.3 Inter-core boot and service flow

Stage	M7 action	M4 action
1	Configure MPU/cache and initialize HAL/clock	Wait in STOP mode
2	Release HSEM0	Wake and clear HSEM event
3	Initialize shared-ring producer	Initialize shared-ring consumer
4	Create OpenAMP master endpoint	Create OpenAMP remote endpoint
5	Produce records; poll messages	Consume records; service messages

MotionIPC Shared-Memory Map

ST OpenAMP resources remain separate from the custom high-throughput motion-data path



Engineering review finding 4,096 × 64-byte records plus metadata exceed the 256 KiB allocation by approximately 64 bytes.

Figure 3. Shared-memory layout and the source-derived capacity review finding.

4. IPC Protocol and Data Structures

4.1 Control-plane message set

Message	Purpose
HEARTBEAT_REQ / REPLY	Per-core liveness, record counts, gaps, and drops
RING_STATUS_REQ / REPLY	Read/write indices, samples read, last sequence, and overflow statistics
START_SEGMENT / STOP_SEGMENT	Worker-state transition hooks and end-of-segment reporting
CLEAR_STATS	Reset consumer counters for a new run
FAULT	Propagate core-local fault flags

4.2 High-rate motion record

Field	Role
seq	Record sequence number
t_us	Microsecond timestamp
encoder_count	Raw encoder count
position_mdeg	Position in millidegrees
target_speed_mdeg_s	Commanded speed in millidegrees per second
measured_speed_mdeg_s	Measured speed in millidegrees per second
error_mdeg	Position or tracking error in millidegrees
state / flags	Compact target-state and condition indicators

WHY TWO IPC PATHS

RPMsg is used where message semantics matter. The ring is used where deterministic, low-overhead record transfer matters. This avoids paying RPMsg framing and endpoint-service costs for every sample.

4.3 Throughput

At 100 records/s and 64 bytes/record, the custom shared-memory path moves 6.4 kB/s (51.2 kb/s), or approximately 384 kB/min. Regular control traffic is only several hundred application bytes per second before RPMsg framing, so the communication architecture preserves substantial headroom.

5. Concurrency, Coherency, and Reliability

Mechanism	Engineering effect
Sequence continuity	Monotonic record numbers and CM4 gap counting expose missing data.
Overflow visibility	Producer increments a drop counter when the ring cannot accept a record.
Core liveness	Independent M7 and M4 heartbeat counters distinguish core-local failures.
Fault propagation	Each core maintains fault flags; RPMsg fault messages transfer status.
Bounded worker load	CM4 consumes no more than 128 records per pass, preserving OpenAMP service cadence.
Memory ordering	Shared-ring implementation uses barriers around producer/consumer state.
Cache safety	M7 MPU policy marks the AXI ring region non-cacheable for coherency.

5.1 Capacity review finding

CORRECTIVE ACTION

The configured $4,096 \times 64$ -byte sample array consumes the full 256 KiB allocation before ring metadata is counted. The public report therefore records this as an engineering review finding. The clean correction is to reduce configured capacity to 4,095 entries or enlarge the reserved memory region.

5.2 Maturity boundary

The supplied source set verifies the dual-core IPC architecture and also contains the target motion-control, telemetry, GUI, and characterization stack. The reviewed archive does not provide one release binary proving every module was merged into a single final build; public claims therefore distinguish verified architecture modules from a fully release-qualified integrated binary.

6. Target Motion Firmware

The target-side motion application keeps pulse timing, encoder capture, test sequencing, and safety behavior on the embedded controller rather than relying on desktop timing. The implementation uses microsecond timing, a fractional pulse accumulator, bounded correction, and target-side health monitoring.

Control/safety item	Verified value
STEP pulse high time	4 μ s
Step-rate safety cap	25,000 pulses/s
Maximum pulse backlog emitted per pass	25 pulses
Loop-overflow warning threshold	25 ms
Speed-estimation window	500 ms
Position-correction gain	20 pulse-Hz/degree
Absolute correction cap	250 Hz
Relative correction cap	5% of feedforward rate
Host heartbeat	500 ms
Target deadman timeout	2,000 ms

6.1 Encoder and motion scaling

Parameter	Value
Motor full steps/revolution	200
Microsteps/full step	256 in ultra-slow configuration
Gear ratio	24.65:1
Driver pulses/table revolution	1,262,080
Pulse command increment	$\sim 0.000285^\circ/\text{pulse}$
Encoder CPR	5,000
Effective table counts/revolution	246,500
Measured angular increment	$\sim 0.001460^\circ/\text{count}$

7. Host Software and External Telemetry

The host application provides operator control, live plotting, data capture, metadata retention, test orchestration, and postprocessing. It is intentionally supervisory: target timing remains on the embedded system.



Figure 4. Host interface overview with connection, test, and live-telemetry controls.

- 115,200-baud serial host interface with selectable 10, 20, 50, or 100 Hz telemetry requests.
- 24-field timestamped CSV records with separate metadata and controller-comment retention.
- Header and row-width validation, malformed-row detection, completion-marker handling, and partial-run recovery.
- Live command-versus-measured speed, encoder position, raw encoder rate, status, and logging state.
- Automated execution and postprocessing for tracking, static hold, step response, Allan deviation, and closed-loop sine response.

7.1 External serial bandwidth review

A 115,200-baud 8-N-1 link provides an idealized 11,520 payload bytes/s. Full 24-field ASCII records can approach roughly 189 bytes each, so a requested 100 Hz stream would require about 18.9 kB/s. The interface therefore supports a 100 Hz selection, but sustainable full-record transmission is closer to approximately 50–60 Hz unless the row is shortened, the link rate is increased, or binary framing is used.

ENGINEERING OUTCOME

The dual-core shared-memory path removes high-rate inter-core data movement from the external serial bottleneck. The serial link remains an operator and logging interface rather than the internal real-time transport.

8. Characterization Evidence

Characterization results are retained as evidence that the firmware and host workflow operated against a physical motion platform. They are supporting evidence—not the primary contribution of this report.

SDC2 ONLY | 256 MICROSTEP BASELINE

Speed tracking summary: usable ultra-slow range, clear high-speed ceiling

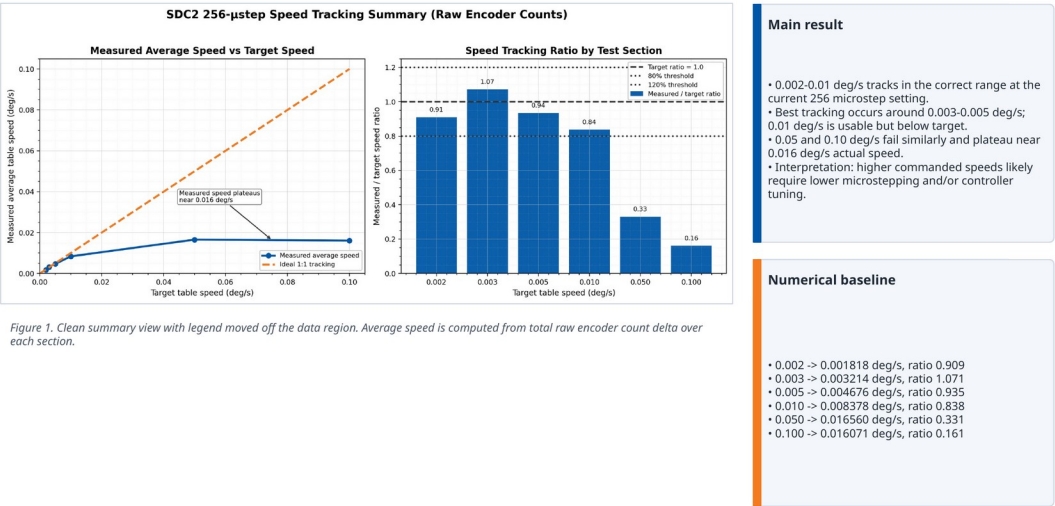


Figure 1. Clean summary view with legend moved off the data region. Average speed is computed from total raw encoder count delta over each section.

SDC2 Encoder-based Speed Tracking

Raw encoder-event analysis | no filtering or grouped averaging

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Figure 5. Configuration-specific low-speed tracking summary for the 256-microstep mode.

Command (deg/s)	Measured (deg/s)	Tracking ratio / note
0.002	0.001818	0.909
0.003	0.003214	1.071
0.005	0.004676	0.935
0.010	0.008378	0.838
0.050	~0.016	Throughput ceiling
0.100	~0.016	Throughput ceiling

8.1 Test and acquisition capabilities

Capability	Configured scope
Static acquisition	50 Hz × 180 s or 100 Hz × 90 s; 9,000 records
Dynamic segment	100 Hz requested; 6,000 records per 60 s segment
Closed-loop sine tests	Approximately 0.33–2 Hz
Safety supervision	500 ms host heartbeat; 2 s target-side deadman

9. Public Deliverables and Source Boundary

- This firmware/software FRD in PDF and DOCX formats.
- Dual-core architecture, memory-map, and verified-specification graphics.
- Selected owned source excerpts defining protocol, memory map, and shared-ring records.
- Host-target demonstration video and representative GUI captures.
- Sanitized characterization plots and configuration-specific low-speed data.
- Complete private development archives, credentials, local paths, and duplicate historical packages excluded.

10. Limitations and Open Validation

Item	Controlled statement
Integrated release binary	Architecture and application modules are source-verified; a single final merged binary is not included in the reviewed evidence package.
Ring allocation	Configured 4,096-entry ring requires a capacity or memory-reservation correction.
IPC latency/jitter	No instrumented end-to-end latency or worst-case jitter result is present.
Serial throughput	100 Hz is selectable, but full ASCII rows exceed 115,200-baud sustainable throughput.
Power	Total electrical power and peak-current measurements are not present.
Reliability percentage	Mechanisms are implemented, but no MTBF or percentage improvement is claimed.

11. Attribution and Revision Control

Herder Elektronische Systemen developed the public architecture, firmware, host software, analysis workflow, interface electronics, technical documentation, and curated case-study package represented here. No third-party affiliation or endorsement is claimed.

Rev	Date	Description
2.3	2026-08-03	Final layout-refined firmware-first release with verified STM32H747/OpenAMP architecture, memory map, IPC rates, timing constants, coherency controls, and public source excerpts.

RELEASE GATE

Before a production release, correct the ring allocation, integrate and tag the final build, instrument IPC latency and jitter, and validate sustained host throughput.